

Effect of dissolved inorganic salts on the isothermal vapor–liquid equilibrium of the propionic acid–water mixture

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Abstract

Propionic acid–water is one of the minimum boiling point azeotropic binary mixtures. Alteration of the vapor–liquid equilibrium (VLE) by addition of inorganic salts is of interest, since it allows bypassing the azeotropic point due to salt-in and salt-out effects. The effect of NaCl, NH₄Cl, CaCl₂, and AlCl₃ at salt concentrations of 1 m and the effect of NaCl molality (0.5–3 m) on the VLE of the binary propionic acid–water mixture has been experimentally investigated at 60 °C using the headspace gas chromatography (HSGC) technique. The alteration in the VLE of the system was salt type and salt concentration dependent. Salting out of the propionic acid was observed when salts were added and was most obvious in the case of AlCl₃ salt. The salt effect parameter, determined from the Furter's equation was a function of the liquid concentration.

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1. Introduction

Distillation is one of the most important techniques used in the separation of liquid mixtures. Distillation, in some cases, however, is not the most appropriate technique especially for liquid mixtures with low relative volatility, or when azeotropes are formed. To circumvent this difficulty the azeotropic point is usually shifted or broken by adding a mass-separating agent such as in azeotropic or extractive distillation or by altering column pressure. In both cases an extra column for pressure adjustment or solvent recovery is needed which adds cost to the process. Recently, extractive distillation employing a dissolved salt rather than a liquid as a separating agent is gaining attention. An advantage of using salt rather than liquid as a separating agent is that salt would not be present in the distilled product since it is a non-volatile and, therefore, no extra separating unit is needed.

When a salt is added to an azeotropic mixture, it may affect the boiling point, the mutual solubilities of the liquid components, and, therefore, the equilibrium vapor phase composition due to salt in and salt out effects. The component of which the equilibrium vapor is enhanced is said to have been salted out and the other component is salted in. There are different approaches for interpreting the salt effect such as hydration of salt ions in the solution, internal pressure effect, electrostatic theory, van der Waals force, and preferential association [1]. Usually, the electrostatic field of the ions preferentially attracts the molecules of the more polar component and hence the vapor composition is enriched by the less polar component, in which the salt is less soluble [2].

Many research papers concerning the salt effect on vapor–liquid equilibrium (VLE), especially the alcohol–water systems, were published [3–8]. Most VLE data available in the literature were obtained either by static still methods or recirculation still methods. The main disadvantages of circulation methods involve the difficulty in obtaining steady state conditions, especially for systems of large relative volatilities where equilibrium is almost impossible to establish, and the difficulty in analyzing the samples of the phases accurately [9]. On

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the other hand, the main problem in static still methods is the need to degas the components, which requires carefully designed apparatus.

An alternative method for the determination of VLE data is the static headspace gas chromatography (HSGC). The main features of this method are reliability, fastness, accuracy, and feasibility of automating the whole procedure. The first application of automated static HSGC was the quantitative determination of ethanol in blood samples. Thereafter, this method was applied in the study of fluid phase equilibria including determining the activity coefficients of components in solution for VLE and liquid–liquid equilibria (LLE) studies [10] and analyzing compositions of headspace vapor phase at equilibrium for studying the effect of porous media on the vapor and liquid equilibria [11]. The principle of this technique is based on the gas chromatographic analysis of the sample gas vapor in thermodynamic equilibrium with the sample in closed vial. Under these conditions, the quantity of volatile components in the headspace gas is proportional to their concentrations in the sample.

For the design of separation processes involving salt components, an accurate description of the VLE behavior is necessary. Some results concerning salt effect on VLE of the propionic acid–water system using the traditional VLE measuring techniques are available in the literature [12]. The objective of the present investigation is to examine the practical feasibility of using the static HSGC technique in measuring the VLE of propionic acid–water system. Furthermore, the effect of inorganic salts such as NaCl, NH_4Cl , CaCl_2 and AlCl_3 on the VLE of propionic acid aqueous solution near the azeotropic point is to be investigated.

2. Experimental

The system utilized in the present study, included a Varian 3400 GC with flame ionization detector, a PC based integrator and an automated static headspace autosampler.

In the experiments, 3 ml of liquid samples of known concentrations were transferred via a pipette to the 22 ml Pyrex glass vials. The vials were immediately sealed with Teflon faced silicone septa and secured by aluminum caps. The vials were then mounted into the carousel of a headspace autosampler. A Genesis headspace autosampler consisting of a carousel, a heated platen, a control panel and a septum needle adapter connected to a sample loop, was used for the sampling. The carousel can hold up to 50 vials, and a maximum of 12 vials can be simultaneously heated in the platen. The vials were heated at constant temperature in the platen until equilibrium between gas and liquid phases has been established. The equilibrium time was deter-

Table 1
Summary of autosampler parameters

Model	Genesis headspace autosampler with 50-position carousel
Carrier gas	Ultra High Purity Helium (Matheson)
Carrier gas flow rate	15 cm^3/min
Platen temperature	60 $^\circ\text{C}$
Equilibrium time	35 min
Sample loop volume	10 μl
Line and valve temperature	175 $^\circ\text{C}$
Loop fill time	0.03 min

mined experimentally by using vials containing 20 mol% of propionic acid–water mixture heated in the platen for a certain period of time, and plotting the area count of the propionic acid peak obtained from the GC as a function of time spent on the platform. Once the VLE was reached, each vial was raised onto the needle, and a fixed volume of the vapor sample (the headspace) was automatically withdrawn. The sample in the loop was then sent to the GC for analysis by Helium carrier gas. A summary of the autosampler parameters is shown in Table 1.

The vapor sample sent from the autosampler was analyzed by a Varian 3400 GC and the Analog-to-Digital converter (ADC) board controlled by a computer using the software Star Workstation. The gas chromatography was calibrated by measuring responses for a set of ten standard solutions with known composition that were prepared gravimetrically. A calibration curve converting peak areas to the propionic acid mole fraction in the vapor phase was generated following the method first proposed by Wong [13] and later confirmed by Shin [11,14]. A detailed description of the calibration method can be found in the above references. A summary of GC parameters used in the experiments is displayed in Table 2.

Table 2
Summary of GC parameters

Model	Varian 3400 GC
Column	DB-624 glass capillary column
Column temperature	80 $^\circ\text{C}$
Ramping temperature	180 $^\circ\text{C}$
Detector	Flame Ionization Detector (FID)
Detector temperature	250 $^\circ\text{C}$
Injector temperature	160 $^\circ\text{C}$
Make-up gas	Ultra High Purity Helium
Make-up gas flow rate	15 cm^3/min
Air flow rate	300 cm^3/min
Hydrogen gas flow rate	30 cm^3/min

3. Chemicals

Distilled water and propionic acid of 99.9% purity (Sigma) were used. The salts used in this study had purities of 99.5+ %.

4. Results and discussion

The VLE data for the binary propionic acid–water system, in the absence of salt, obtained in this work was firstly compared with the literature data at 60 °C to verify the practical feasibility of using the HSGC technique for studying the VLE of binary organic acid–water mixtures. It is seen (Fig. 1) that the data of this work are almost concurring the literature ones. This would indicate that the experimental apparatus and procedures used in this study give precise and reproducible data.

The VLE data for the propionic acid–water system at 60 °C in the presence of 1.0 m of NaCl or CaCl₂ or AlCl₃ or NH₄Cl are displayed in Table 3 and are shown in Fig. 1. The results (Fig. 1) show that the VLE data of propionic acid–water system is altered in the presence of any of these salts. In addition, the azeotropic point of the considered system is shifted upward. The relative volatility of propionic acid increased in the presence of salts indicating that these salts have a salting out effect

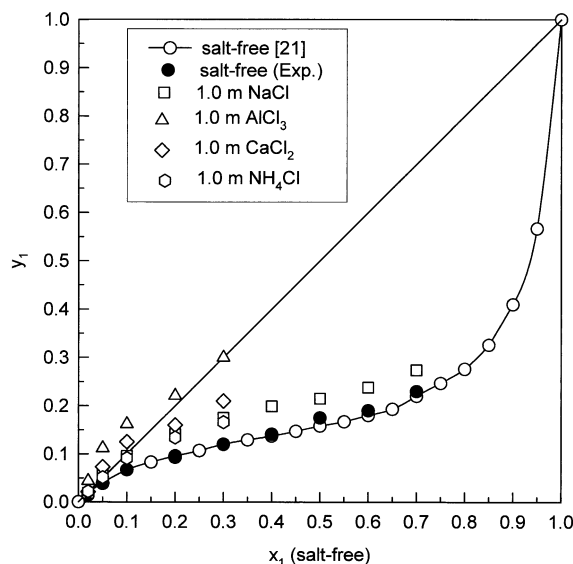


Fig. 1. x – y vapor–liquid equilibrium diagram for the system propionic acid(1)+water (2)+salts (3) at 60 °C.

upon propionic acid. The salting out effect occurs as a result of the formation of salt–liquid phase complexes with one species in the liquid more than the other. The formation of these complexes reduces the activity of the more attracted component in the solution and increases the activity of the less attracted component, which result in expelling or salting out the latter from the liquid

Table 3

Vapor–liquid equilibria data of propionic acid–water–salt systems at 60 °C

x_1	y_1 (salt-free)	y_1 (with 1 m NaCl)	y_1 (with 1 m AlCl ₃)	y_1 (with 1 m CaCl ₂)	y_1 (with 1 m NH ₄ Cl)
0.020	0.0143	0.0185	0.0440	0.0210	0.0220
0.050	0.0383	0.0598	0.1120	0.0730	0.0514
0.100	0.0670	0.0959	0.1620	0.1250	0.0912
0.200	0.0930	0.1410	0.2210	0.1600	0.1340
0.300	0.1200	0.1750	0.3000	0.2100	0.1660
0.400	0.1410	0.1990			
0.500	0.1750	0.2150			
0.600	0.1900	0.2380			
0.700	0.2300	0.2740			

Table 4

Vapor–liquid equilibria data of propionic acid–water–NaCl salt systems at 60 °C

x_1	y_1 (salt-free)	y_1 (with 0.5 m NaCl)	y_1 (with 1 m NaCl)	y_1 (with 2 m NaCl)	y_1 (with 3 m NaCl)
0.020	0.0143	0.0142	0.0185	0.0291	0.0466
0.050	0.0383	0.0495	0.0598	0.0871	0.1170
0.100	0.0670	0.0828	0.0959	0.1280	0.1620
0.200	0.0930	0.1180	0.1410	0.1710	0.1950
0.300	0.1200	0.1530	0.1750	0.1810	
0.400	0.1410		0.1990		
0.500	0.1750		0.2150		
0.600	0.1900		0.2380		
0.700	0.2300		0.2740		

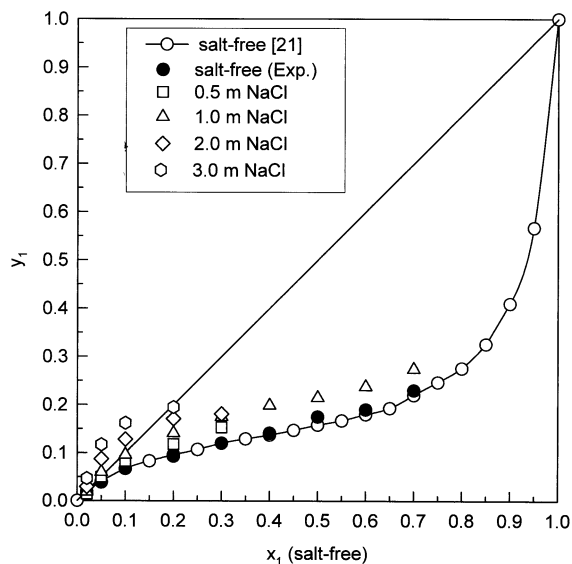


Fig. 2. Effect of NaCl concentration on the x - y diagram of propionic acid–water system at 60 °C.

solution, hence increasing its relative volatility. However, such an effect is different from one salt to another; the relative volatility of propionic acid in the presence of 1.0 m AlCl_3 is greater than that in the presence of 1.0 m CaCl_2 followed by 1.0 m NaCl and finally by 1 m NH_4Cl . It seems that there is a correlation between the ionic radius and the salting out effect, the salting out effect increases with decreasing atomic radius. This observation is consistent with the electrostatic theory

and with the order of effectiveness observed by previous investigators [15,16]. The ionic radii for aluminum, calcium and sodium are 0.053, 0.100 and 0.102 nm, respectively. The smaller ion, allows a closer approach to its charge center, which in turns exerts a stronger attractive force for molecular association in solution.

The effect of salt concentration on the salting out effect was investigated using NaCl with different molalities, 0.5, 1.0, 2.0 and 3 m. The solubility of NaCl in water at 60 °C is around 6 m whereas it is almost insoluble in propionic acid, with a solubility being no more than 0.002 m [17]. The experimental results are presented in Table 4 and are shown in Fig. 2. The VLE data (Fig. 2) demonstrates that the salting out effect increases with increasing salt concentration. This trend was expected, since the increase in salt concentration increases the intermolecular forces between the salt and the polar component, water, in the mixture. Thus, the salting out effect of the component of less polarity, propionic acid, increases as the salt molality increases.

For the propionic acid–water mixture, the propionic acid monomers associate with each other to form hydrogen-bonded dimers in both liquid and vapor phases. Due to dimerization the system deviates even from the ideal vapor phase behavior. For example, the vapor phase compressibility factor (Z) of propionic acid at a pressure of 0.016 bar and temperature of 40 °C is 0.58 [18]. As is evident, even though the pressure is very low, the vapor compressibility factor deviates considerably from unity, when in fact we would expect the ideal

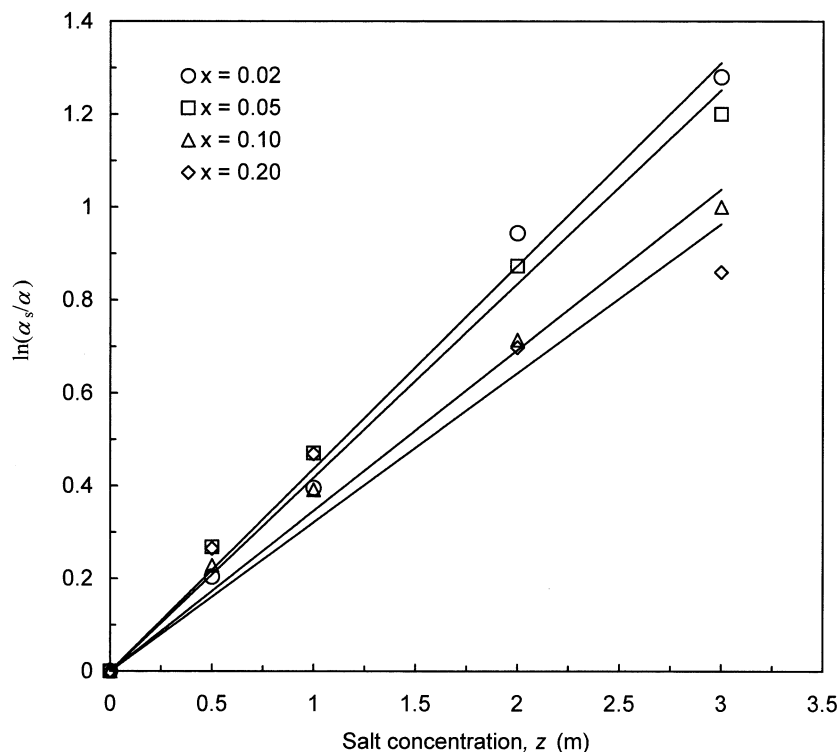


Fig. 3. Salt effect on propionic acid–water system at 60 °C with NaCl.

Table 5

Salt effect parameters and empirical model parameters at different propionic acid concentration

Propionic acid (x_1)	Salt effect (k)
0.02	0.4368
0.05	0.4175
0.10	0.3462
0.20	0.3212

gas law to hold and $Z = 1$. Therefore, to account for the dimerization effect, the experimental activity coefficients, γ_1 and γ_2 , can be calculated by applying the chemical theory model [19]. However, because the application of the chemical theory model requires precise measurement of the VLE pressure, which was not possible in the conducted experiments, this model will not be used here.

The effect of salt on the VLE can be seen more easily from the ratio of the relative volatilities [20]:

$$\ln\left(\frac{\alpha_s}{\alpha}\right) = kz \quad (1)$$

where α_s and α are relative volatilities in the presence of salt and in the absence of salt, respectively, k is a salt effect parameter that depend on the system and solvent composition, and z is the salt concentration (m). Eq. (1) is called Furter' equation. Fig. 3 shows the relationship

Table 6

Constants m and c of Equation 8 at different NaCl concentrations

Salt concentration (m)	m	c
0.5	0.942	0.2107
1.0	0.9285	0.3947
2.0	1.299	0.9274
3.0	1.550	1.3117

between $\ln(\alpha_s/\alpha)$ versus salt concentration, z . The values of k , slope of the straight lines in Fig. 3, are shown in Table 5. From this table and from the lines of Fig. 3 it is seen that the salt effect parameter, k , decreases with increasing propionic acid concentration. When the solvent composition is fixed, the salt effect is proportional to the salt concentration.

The effect of salt on the relative volatility can also be analyzed using the following correlation [5] for each salt concentration:

$$\ln \alpha_s = m \ln \alpha + c \quad (2)$$

According to Eq. (2) a plot of $\ln \alpha_s$ versus $\ln \alpha$ at a particular salt concentration (Fig. 4) gives a straight line with slope m and intercept c . These constants, m and c , characterize the system under consideration, their numeric values are 1.0 and zero, respectively, in the absence of salt. The m and c values are displayed in Table 6 at different salt concentrations. It is seen, Fig. 4

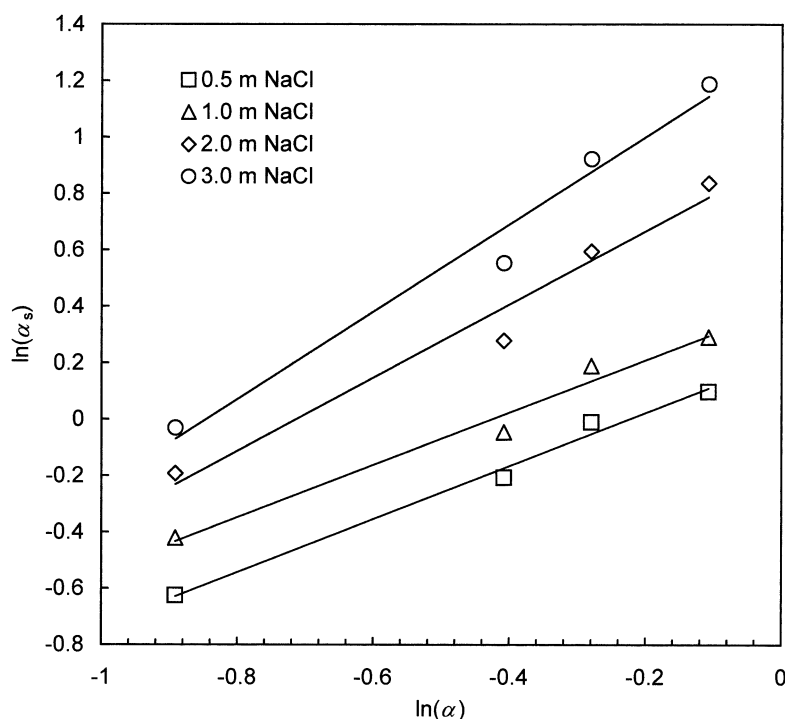


Fig. 4. Plots of $\ln \alpha_s$ against $\ln \alpha$ for propionic acid–water system at 60 °C and different salt concentrations.

and Table 6, that m and c are salt concentration dependent, and that they deviate far from 1.0 and zero, respectively, with increasing salt concentration, indicating the influence of salt on the VLE behavior.

5. Conclusions

The effect of the salts sodium chloride, calcium chloride, ammonium chloride, and aluminum chloride on the VLE of the propionic acid–water mixture has been experimentally investigated using a HSGC technique. All salts investigated in this work exhibited a salting out effect on the propionic acid. The salting out effect of NaCl and NH_4Cl was almost identical and less than the effect of the other two salts. AlCl_3 was the most effective salt. The degree of alteration in the VLE was found to increase with increasing NaCl salt molality. The salt effect parameter obtained by fitting the experimental data to the Furter's equation was dependent on the liquid composition.

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